



Gives answers one after other
iam having train.csv file



Perfect 👍 — since you already have **train.csv**, below is a **step-by-step Python solution** that follows your lab manual **in exact order (Week-5 Decision Tree)**.

You can copy-paste this into Jupyter Notebook or Google Colab and run **cell by cell**.

✅ WEEK-5 : DECISION TREE (Titanic Dataset)

◆ 1. Import Necessary Libraries

python

 Copy code

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

◆ 2. Load Dataset (train & test)

python

 Copy code

```
train = pd.read_csv("train.csv")
test = pd.read_csv("train.csv") # if you don't have separate test file
```

◆ 3. Look at Top Values of Train Data

python

 Copy code

```
train.head()
```

◆ 4. Print Column Names

python

 Copy code

```
train.columns
```

◆ 5. Check Data Types

python

 Copy code

```
train.dtypes
```

◆ 6. Print Dimensions

python

 Copy code

```
print("Train Shape:", train.shape)
print("Test Shape:", test.shape)
```

◆ 7. Summary Statistics

python

 Copy code

```
train.describe()
```

◆ 8. Missing Values

python

 Copy code

```
train.isnull().sum()
```

◆ 9. Bar Chart (survived, pclass, sex, embarked)

python

 Copy code

```
plt.figure(figsize=(12,6))

plt.subplot(2,2,1)
sns.countplot(x="Survived", data=train)

plt.subplot(2,2,2)
sns.countplot(x="Pclass", data=train)

plt.subplot(2,2,3)
sns.countplot(x="Sex", data=train)

plt.subplot(2,2,4)
sns.countplot(x="Embarked", data=train)

plt.tight_layout()
plt.show()
```

◆ 10. Histogram of Age (Survived vs Not Survived)

python

 Copy code

```
plt.hist(train[train["Survived"]==1]["Age"], alpha=0.5, label="Survived")
plt.hist(train[train["Survived"]==0]["Age"], alpha=0.5, label="Not Survived")
plt.legend()
plt.xlabel("Age")
plt.ylabel("Count")
plt.show()
```

◆ 11. Replace Missing Embarked with 'S'

python

```
train["Embarked"].fillna("S", inplace=True)
```

◆ 12. Impute Fare with Median

python

 Copy code

```
train["Fare"].fillna(train["Fare"].median(), inplace=True)
```

◆ 13 & 14. Impute Age Using Mean & Std

python

 Copy code

```
mean_age = train["Age"].mean()
std_age = train["Age"].std()
```

```
rand_age = np.random.randint(mean_age-std_age, mean_age+std_age, size=train.
train.loc[train["Age"].isnull(), "Age"] = rand_age
```



◆ 15–16. Create Family Size

python

 Copy code

```
train["FamilySize"] = train["SibSp"] + train["Parch"] + 1
```

◆ 17–18. Create IsAlone

python

 Copy code

```
train["IsAlone"] = 0
train.loc[train["FamilySize"]==1, "IsAlone"] = 1
```

◆ 19–20. Drop Columns

python

 Copy code

```
train.drop(["PassengerId", "Name", "Ticket", "Cabin"], axis=1, inplace=True)
```

◆ 21. Print Top 5 Rows

python

 Copy code

```
train.head()
```

◆ 22. Pairplot

python

 Copy code

```
sns.pairplot(train)
plt.show()
```

◆ 23. Prepare X and Y

python

 Copy code

```
X = train.drop("Survived", axis=1)
Y = train["Survived"]
```

◆ Convert Categorical to Numeric

python

 Copy code

```
X = pd.get_dummies(X, drop_first=True)
```

◆ 24. Train-Test Split

python

 Copy code

```
from sklearn.model_selection import train_test_split
```

```
X_train, X_val, Y_train, Y_val = train_test_split(X, Y, test_size=0.2, rand
```



◆ 25–26. Create & Fit Decision Tree

python

 Copy code

```
from sklearn.tree import DecisionTreeClassifier
```

```
dt = DecisionTreeClassifier()  
dt.fit(X_train, Y_train)
```

◆ 27. Print Tree

python

 Copy code

```
from sklearn.tree import plot_tree
```

```
plt.figure(figsize=(15,10))  
plot_tree(dt, filled=True)  
plt.show()
```

◆ 28–32. Evaluation

python

 Copy code

```
from sklearn.metrics import confusion_matrix, accuracy_score, classification_report
```

```
pred = dt.predict(X_val)
```

```
print("Accuracy:", accuracy_score(Y_val, pred))  
print(confusion_matrix(Y_val, pred))  
print(classification_report(Y_val, pred))
```



◆ 33–38. GridSearchCV

python

 Copy code

```
from sklearn.model_selection import GridSearchCV

params = {
    "max_depth": [3, 5, 7, 10],
    "min_samples_split": [2, 5, 10],
    "min_samples_leaf": [1, 2, 4]
}

grid = GridSearchCV(DecisionTreeClassifier(), params, cv=5)
grid.fit(X_train, Y_train)

grid.best_params_
```

◆ 39. Plot Accuracy vs Max Depth

python

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```
depths = [3, 5, 7, 10]
scores = []

for d in depths:
    model = DecisionTreeClassifier(max_depth=d)
    model.fit(X_train, Y_train)
    scores.append(model.score(X_val, Y_val))

plt.plot(depths, scores)
plt.xlabel("Max Depth")
plt.ylabel("Accuracy")
plt.show()
```

◆ 40–42. Optimal Model

python

 Copy code

```
best_dt = DecisionTreeClassifier(**grid.best_params_)
best_dt.fit(X_train, Y_train)
```

best_dt

◆ 43–47. Final Evaluation

python

 Copy code

```
final_pred = best_dt.predict(X_val)

print("Accuracy:", accuracy_score(Y_val, final_pred))
print(confusion_matrix(Y_val, final_pred))
print(classification_report(Y_val, final_pred))
```

✓ YOU ARE DONE 🎉

This is **lab-ready**, **viva-safe**, and **exam-oriented**.

If you want, I can also give:

- ✓ Viva Questions & Answers
- ✓ Short theory notes
- ✓ Naive Bayes (Week-6) full code

Just tell me 👍

     ...

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

Great 👍 — you have **successfully completed Step-1 (Importing Libraries)**.

Now let's continue **one after another** exactly as your lab manual says.

✅ Step-2: Load the Dataset (train.csv)

Make sure your `train.csv` file is in the same folder as your notebook.

python

 Copy code

```
train = pd.read_csv("train.csv")
test = pd.read_csv("train.csv")    # using same file as test
```

✅ Step-3: Look at Top Values of Train Data

python

 Copy code

```
train.head()
```

✅ Step-4: Print Column Names

python

 Copy code

```
train.columns
```

✅ Step-5: Check Data Type of Each Column

python

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```
train.dtypes
```

✅ Step-6: Print Dimension of Train and Test Data

python

 Copy code

```
print("Train Shape:", train.shape)
print("Test Shape:", test.shape)
```

✅ Step-7: Summary Statistics of Train Data

python

 Copy code

```
train.describe()
```

✅ Step-8: Missing Values in Each Column

python

 Copy code

```
train.isnull().sum()
```

Run these steps first.

After that, tell me **"done"**, then we will move to **bar charts and visualization (Step-9)**

